UNIQ ID: E0124092

```
import gymnasium as gym
import numpy as np


env = gym.make("CartPole-v1")

state, info = env.reset()

print("Action Space:", env.action_space)
print("Observation Space:", env.observation_space)


def describe_state(state):

    cart_position = state[0]
    cart_velocity = state[1]
    pole_angle = state[2]
    pole_velocity = state[3]

    print("Cart Position:", cart_position)
    print("Cart Velocity:", cart_velocity)
    print("Pole Angle:", pole_angle)
    print("Pole Velocity:", pole_velocity)


print("\nInitial State:")
describe_state(state)



actions = {
    0: "Move Left",
    1: "Move Right"
}

print("\nActions:")

for action in actions:
    print(f"Action {action}: {actions[action]}")


num_steps = 5

print("\nSimulating a few steps:")

for step in range(num_steps):

    action = env.action_space.sample()

    next_state, reward, terminated, truncated, info = env.step(action)

    done = terminated or truncated
```

```
    print(f"\nStep {step + 1}:")
    print("Action taken:", actions[action])

    print("Next State:")
    describe_state(next_state)

    print("Reward:", reward)
    print("Done:", done)

    state = next_state

    if done:
        print("\nEpisode ended. Resetting environment...")
        state, info = env.reset()

env.close()
```

```
Action Space: Discrete(2)
Observation Space: Box([-4.8               -inf -0.41887903        -inf], [4.8
inf 0.41887903        inf], (4,), float32)

Initial State:
Cart Position: -0.040507488
Cart Velocity: -0.042971075
Pole Angle: 0.0012500514
Pole Velocity: -0.03216326

Actions:
Action 0: Move Left
Action 1: Move Right

Simulating a few steps:

Step 1:
Action taken: Move Right
Next State:
Cart Position: -0.04136691
Cart Velocity: 0.15213293
Pole Angle: 0.0006067862
Pole Velocity: -0.3244515
Reward: 1.0
Done: False

Step 2:
Action taken: Move Right
Next State:
Cart Position: -0.03832425
Cart Velocity: 0.34724623
Pole Angle: -0.005882244
Pole Velocity: -0.616943
Reward: 1.0
Done: False

Step 3:
```

```
Action taken: Move Left
Next State:
Cart Position: -0.031379327
Cart Velocity: 0.15220696
Pole Angle: -0.018221105
Pole Velocity: -0.32611853
Reward: 1.0
Done: False

Step 4:
Action taken: Move Right
Next State:
Cart Position: -0.028335188
Cart Velocity: 0.34758353
Pole Angle: -0.024743475
Pole Velocity: -0.62449145
Reward: 1.0
Done: False

Step 5:
Action taken: Move Left
Next State:
Cart Position: -0.021383516
Cart Velocity: 0.1528156
Pole Angle: -0.037233304
Pole Velocity: -0.33970284
Reward: 1.0
Done: False

/usr/local/lib/python3.13/dist-packages/jupyter_client/session.py:203:
DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for
removal in a future version. Use timezone-aware objects to represent datetimes
in UTC: datetime.datetime.now(datetime.UTC).
  return datetime.utcnow().replace(tzinfo=utc)
```